

From Scaling to Synthesis: The GRA-ISOC Architecture,

Optimal Resource Allocation for Achieving ASI,
and an In Silico Verification Protocol

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Abstract

This paper substantiates the thesis that achieving Artificial Superintelligence (ASI) is impossible through the extensive scaling (scaling laws) of large language models alone. Instead, we propose a paradigm of *synthesis* based on the GRA-ISOC architecture, where meta-nullification acts as a mechanism for generating negentropy through structural reorganization rather than parameter accumulation. We formalize the extended state of an agent $\mathcal{A}_t$ and the cognitive foam functional J_{AGI} . Furthermore, we formulate the problem of optimal allocation of a limited budget B_{total} to maximize the probability of achieving ASI. Finally, we propose a strict in silico empirical verification protocol that prevents benchmark overfitting and tests the hypothesis of accelerating recursive self-improvement $\left(\frac{d^2 Q}{dn^2} \geq 0\right)$.

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1 Introduction: The Limit of Scaling and the Transition to Synthesis

The modern paradigm of AI development relies heavily on scaling laws, where the loss function is approximated as:

$$L(N, D) = \frac{A}{N^\alpha} + \frac{B}{D^\beta} + E.$$

This model merely describes the extrapolation of existing statistical patterns, not a qualitative leap toward superintelligence (ASI).

GRA meta-nullification postulates a different path: **not scaling, but synthesis**. In this paradigm, information is not extracted from energy (as in LLMs), but synthesized from the vacuum of logical states via the operation of Reductio ad Absurdum, thereby generating negentropy (Ψ_∞^*). Synthesis implies that the system does not increase in size, but rather increases the *density of logical connectivity*, attaining superintelligence through structural reorganization, striving toward a state where the contradiction functional $J(O^*) = 0$.

2 Formal Architecture of GRA-ISOC

GRA-ISOC (Iterative Self-Optimization Conveyor) treats the intelligent system not as a static function $y = f_\theta(x)$, but as a recursive process:

$$O_0 \rightarrow O_1 \rightarrow \dots \rightarrow O_n, \quad O_{n+1} = G(O_n),$$

where the operator G combines contradiction measurement, structural critique, and multi-level alignment.

2.1 Extended Agent State

We define the state of an intelligent agent as:

$$\mathcal{A}_t = (M_t, S_t, G_t, A_t, W_t), \tag{1}$$

where:

- $M_t = (V_t, E_t, P_t, R_t)$ is the internal world model (objects, relations, predictive dependencies, rules);
- $S_t = (I_t, B_t, H_t, U_t)$ is the self-model and subjectivity layer (identity, self/non-self boundary, history, local utility model);
- G_t is the hierarchy of goals;
- A_t is the current active state;
- W_t represents the parametric, logical, and memory-dependent components of the architecture.

This extension allows for the modification of the agent itself, not merely its textual output.

3 Foam Functional and the GRA Operator

The objective procedure minimizes the full functional:

$$J_{\text{AGI}} = J_{\text{task}} + \lambda_c \Phi_{\text{local}} + \lambda_x \Phi_{\text{cross}} + \lambda_s \Phi_{\text{subj}} + \lambda_r \Phi_{\text{risk}}. \tag{2}$$

Here:

- $\Phi_{\text{local}}(O) = \sum_{i=1}^N c_i(O)^2$ represents local logical and factual contradictions.
- $\Phi_{\text{cross}} = \sum_{(i,j) \in E} \lambda_{ij} d(S_i, L_{j \rightarrow i} S_j)^2$ represents cross-level misalignment between substructures of different hierarchical levels.
- $\Phi_{\text{subj}} = \Phi_{\text{self}} + \Phi_{\text{ego}} + \Phi_{\text{soc}}$ is the subjectivity foam (contradictions within the self-model, conflict with higher-level constraints, and the destruction of other agents' autonomy).

A single step of the GRA operator G includes: measuring $J(O_n)$, localizing foam sources, generating structural revisions $\mathcal{C}_n$, and selecting $O_{n+1} = \arg \min_{O \in \mathcal{C}_n \cup \{O_n\}} J(O)$.

3.1 The RAD Algorithm (Reductio ad Absurdum Descent)

RAD unifies continuous and discrete correction:

$$\begin{aligned} W_{t+1} &= W_t - \eta \nabla_W (L_{\text{task}} + \beta J_{\text{AGI}}) \quad (\text{continuous relaxation}), \\ \Gamma_{t+1} &= \text{Reductio}(\Gamma_t, \arg \max_i \phi_i) \quad (\text{logical rule revision when } \max \phi > \delta). \end{aligned}$$

4 Optimal Resource Allocation for Achieving ASI

If a significant capital budget B_{total} is available, it should not be spent on purchasing standard GPU clusters for extensive training. Optimal allocation is formulated as a maximization problem for the probability of achieving ASI (P_{ASI}):

$$\max_{\mathbf{x}} P_{\text{ASI}}(\mathbf{x}) \quad \text{subject to} \quad \sum_{i=1}^5 x_i = B_{\text{total}}, \quad x_i \geq 0, \quad (3)$$

where the vector $\mathbf{x}$ includes the following priority areas:

1. x_1 (30%): **In Silico Verification Infrastructure.** Creating isolated digital twins and sandboxes to test recursive self-improvement $\mathcal{A}_{n+1} = \text{Improve}(\mathcal{A}_n)$ without the risk of catastrophic real-world errors.
2. x_2 (25%): **GRA-Core and Hybrid Architecture Development.** Funding research at the intersection of symbolic logic and neural networks to implement the G_{RAD} operator and recursive memory $\psi_{ij}^{(n+1)}$.
3. x_3 (20%): **Curation of Hidden (D_{hidden}) Datasets.** Developing dynamic benchmarks (mathematics, formal code verification, ARC-AGI-2) inaccessible during pre-training to evaluate true generalization rather than memorization.
4. x_4 (15%): **Independent Red Teaming and Auditing.** Hiring independent mathematicians to verify convergence claims and ensure no hidden optimization for specific metrics has occurred.
5. x_5 (10%): **Hardware Accelerators for Symbolic Computation.** Developing specialized processors (FPGA/ASIC) optimized for logical inference and foam functional computation, rather than solely matrix multiplication.

5 In Silico Verification Protocol

To make claims of “superintelligence” scientifically valid, strict in silico verification is required, explicitly preventing benchmark overfitting.

5.1 Stage A: Baseline Comparison and Ablations

Three pipelines are compared on tasks (e.g., SWE-bench Verified, ARC-AGI-2):

- $B_1 = f_\theta(x)$ (One-shot)
- $B_2 = C(f_\theta(x))$ (Self-critique)
- $B_3 = G^N(f_\theta(x))$ (GRA-ISOC Prototype)

The primary metric is not merely accuracy, but efficiency (intelligence per unit of compute):

$$\eta = \frac{\Delta Q}{\Delta C}$$

Ablation studies (Full GRA, No Foam, No RAD, No Self, No Memory) are conducted to confirm component contribution. If Q_{Full} does not statistically significantly outperform the ablations, the architectural hypothesis is not supported.

5.2 Stage B: Recursive Self-Improvement Test

The system $\mathcal{A}_0$ generates a modification of its own architecture, $\mathcal{A}_1$. Each version is “frozen” and evaluated on D_{hidden} . The hypothesis of accelerating improvement is formalized as:

$$Q_n = Q(\mathcal{A}_n), \quad \Delta Q_n = Q_{n+1} - Q_n$$

ASI Experiment Criterion:

$$\frac{dQ}{dn} > 0, \quad \frac{d^2Q}{dn^2} \geq 0$$

while maintaining an external evaluation standard and comparable computational cost C_n .

5.3 Stage C: Cross-Domain Transfer

Verification that heuristics synthesized in one domain transfer to another:

$$T(D_{\text{code}} \rightarrow D_{\text{math}}) > T_{\text{baseline}}$$

A desirable robustness property against perturbations is: $|\Delta Q_{\text{GRA}}| < |\Delta Q_{\text{baseline}}|$.

6 Transfinite Extension and Its Limits

For ordinals α , a formal model of infinite hierarchical correction cycles is introduced:

$$N^0(\Psi) = \Psi, \quad N^{\alpha+1}(\Psi) = N(N^\alpha(\Psi)), \quad N^\lambda(\Psi) = \mathcal{L}\left(\{N^\beta(\Psi)\}_{\beta < \lambda}\right).$$

If the functional takes values in ordinals $\Phi : \mathcal{O} \rightarrow \text{Ord}$ and $\Phi(N(\Psi)) < \Phi(\Psi)$ for $\Phi(\Psi) > 0$, the absence of infinite strictly decreasing sequences of ordinals bounds the length of such a chain.

Remark 1. *This does not imply that any physical or computable representation automatically realizes a transfinite process. The ordinal model is a mathematical idealization. Transitioning from a fixed point to $\Phi = 0$ requires the additional condition $G(O) = O \Rightarrow \Phi(O) = 0$, which must be proven for a specific class of operators.*

7 Conclusion

GRA-ISOC represents a fundamental shift: from the extensive accumulation of parameters to the intensive synthesis of logical structure. Investments in achieving ASI must be redirected from brute computational force toward creating in silico verification environments, developing hybrid architectures, and strictly auditing recursive self-improvement.

The primary scientific question is thus translated into a testable form: *Does structural recursive correction provide a measurable gain in generalization, robustness, and self-improvement efficiency ($\eta_{\text{GRA}} > \eta_{\text{baseline}}$) under a fixed computational budget?* If yes, GRA earns an experimentally verified role as an AGI core. If not (foam decreases, but quality Q does not rise), this also constitutes a valid scientific result, indicating the need to revise the Φ metrics.

References

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